



PhysioSync-AR

*Zero-Shot Dynamic Spatial Audio-Visual Keyframe Hashing for Real-Time,
Privacy-Preserving AR Video Summarization*

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Problem Definition Stage

Team Members

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Title Finalization

Finalized Title: “**Zero-Shot Dynamic Spatial Audio-Visual Keyframe Hashing for Real-Time, Privacy-Preserving AR Video Summarization**” (Project Codename: PhysioSync-AR).

The title was chosen to precisely name the four elements that make the method novel and patentable, rather than describing it generically as a “summarization tool”:

- Spatial Audio-Visual: signals both channel-aware audio and video are used jointly, not audio or video alone.
- Keyframe Hashing: identifies the core mechanism as a compact hashing scheme rather than a full deep-learning classifier.
- Zero-Shot / Real-Time: signals the system requires no per-deployment retraining and operates at live (30+ FPS) speed on CPU.
- Privacy-Preserving AR: scopes the application domain to AR/wearable video, and flags that skeletal-hash representations avoid storing raw identifiable imagery.

Abstract

Real-time video summarization for augmented-reality (AR) and wearable-camera streams is typically approached with single-modality heuristics — global colour-histogram shifts for keyframe detection, or raw RMS volume spikes for audio salience. These techniques were designed for flat, single-channel media and break down on spatial, multi-sensor AR footage, where the composition of a frame can stay visually static while the acoustic scene or a person's physical motion changes meaningfully. This proposal introduces PhysioSync-AR, a system built around a novel process — Cross-Modal Spatial Attention Hashing — that fuses inter-channel stereo audio phase differences (sound localization) with human skeletal pose-trajectory vectors into a single 64-bit Spatial-Acoustic Importance Score used to select keyframes. Unlike prior keyframe-extraction and CDN/media-caching approaches, which treat audio and video as independent or content-agnostic signals, this system couples spatial acoustic direction directly to human motion, producing a lightweight, GPU-free, bandwidth-constrained summarization pipeline suited to AR streaming and lecture-capture use cases. The result is a technologically integrated, patentable mechanism rather than an incremental heuristic tweak — positioned for both an Indian Patent Office (IPO) filing and a top-tier multimedia conference submission.

Literature Survey

Section A: Research Papers Survey

Paper / Reference	Problem Addressed	Methodology	Relevance to Project
AudioVisual Video Summarization (AVRN) — Badamdorj et al., IEEE Trans. Multimedia / ResearchGate, 2021	Traditional summarizers focus only on visual cues, ignoring audio cues like speech emphasis or ambient changes.	Two-stream LSTM processes audio and visual streams independently, followed by a fusion LSTM for latent temporal dependencies.	Validates dual-stream (audio+video) models over visual-only ones; this project swaps heavy LSTMs for lightweight ITD phase-difference and pose hashing on CPU.
Multimodal Video Summarization Using ML & Feature Selection — Benchmark Study, Algorithms Journal, 2024/2025	No standard benchmark for low-latency multimodal feature fusion in resource-constrained web/edge environments.	Evaluated pyAudioAnalysis audio features alongside OpenCV visual features (colour histograms, optical flow, object detection) to classify frames.	Direct prior art for combining OpenCV visual metrics with Librosa/SciPy audio metrics for real-time Flask/Streamlit deployment.
VideoXum: Cross-Modal Visual and Textual Summarization — Liu et al., arXiv:2303.12060 / CVPR, 2023	Existing models produce keyframe clips OR text summaries in isolation, losing temporal cross-modal alignment.	VTSUM-BLIP and VT-CLIPScore on ActivityNet jointly produce semantically aligned video clips and narrative summaries.	Shows the need for temporal alignment between keyframes and audio events; this project syncs MediaPipe timestamps with spatial audio energy peaks.
Contrastive Transformer Cross-Modal Hashing	High compute cost and storage overhead when	Bidirectional Transformer encoders with supervised multi-modality contrastive loss	Inspires the lightweight Perceptual Motion & Phase Hashing

Paper / Reference	Problem Addressed	Methodology	Relevance to Project
(CTCH) — ResearchGate Publication, 2024	indexing large-scale multimodal video data.	compress features into compact binary hash codes.	mechanism, enabling real-time matching without Transformer overhead on CPU.
Visual Attention Based Keyframe Extraction — CS & IT Research Publications	Random or fixed-interval sampling misses short critical events or introduces redundancy.	Visual attention metrics (motion vectors, colour dynamics, shot-based clustering) assign saliency scores to frames.	Provides the mathematical basis for dynamic thresholding over uniform sampling in keyframe selection.

Section B: Patents Survey

Patent No. / Assignee	Core Invention	Distinction vs. Project
US10791412B2 (Granted) — Microsoft / Spatial Audio Systems	Determines multi-channel spatial audio properties (position, directional intensity, focus) over time and aligns them visually.	Renders audio as visual particles on a 3D display. This project instead uses directional audio (ITD/phase shift) as a filtering signal to trim and summarize video.
US10166999B1 (Granted) — Vehicle Safety Systems Inc.	Fuses pixel-level video analytics with spatial audio measurements to calculate event locations and trigger surround-sound alerts.	Built for automotive obstacle-alert triggering. This project uses MediaPipe pose vectors and audio phase shifts for multi-criteria keyframe hashing in media summarization.
US10452683B2 / CN110263300A — Baidu / Tencent Tech	Encodes audio and visual feature streams into binary hash keys (Hamming space) for fast multimedia search across databases.	Prior art uses global image/CNN feature vectors. This project uses pose-landmark velocity vectors combined with ITD to generate privacy-preserving skeletal hashes.
KR102145892B1 / US9881640B2 — Samsung Electronics / Qualcomm Inc.	Extracts visual saliency (optical flow) alongside RMS volume spikes to auto-select highlight segments in sports broadcasts.	Relies on monaural RMS audio and dense optical flow. This project replaces dense optical flow with sparse MediaPipe 3D joint trajectories, cutting compute by ~80%.
US11288506B2 (Granted) — Intel Corporation	Uses skeleton estimation to detect high-activity gestures (presentations, sports) and flags keyframes by gesture-velocity thresholds.	Operates on video only. This project adds Cross-Modal Spatial Acoustic Phase Coupling, requiring gesture velocity to align with directional audio changes before a keyframe is selected.

Problem Statement Identification

Video summarization and keyframe-extraction systems in common use rely on one of a small set of well-established heuristics: detecting abrupt changes in global colour histograms, thresholding raw audio volume (RMS energy) for “loud moments,” or running object detectors independently of the audio track. Each of these works reasonably well for flat, single-camera, single-channel video (e.g., broadcast television), but AR and wearable-camera streams are fundamentally different: the camera and microphone array move with the wearer, sound arrives from many directions, and the visual frame can remain largely static while the important event is physical (a person moving, gesturing, turning) or acoustic (someone speaking from off-frame).

Reviewed independently, spatial audio localization, human pose estimation, and lightweight object detection are each mature, well-documented techniques. However, no existing keyframe-extraction system, CDN/media-caching patent, or AR summarization tool combines inter-channel audio phase differences with skeletal pose-trajectory hashing into a single fused importance metric. The table below summarizes where current approaches stop short:

Existing Approach	Limitation
Global colour-histogram keyframing	Detects visual scene changes only; blind to who is speaking, where sound is coming from, or how a person is moving — fails on lecture/AR footage where the frame composition barely changes but the semantic content does.
RMS / volume-spike audio summarization	Flags loudness, not spatial direction; cannot tell that a speaker moved across the room or that an off-screen event is relevant.
Single-modality pose or object detection	Skeletal-motion-only or object-only systems ignore acoustic context entirely, so a quiet but visually static but *acoustically* important moment (e.g., a question from off-frame) is missed.
Generic CDN / media summarization patents	Media-centric and content-agnostic; none couple inter-channel audio phase differences with human pose-trajectory vectors for bandwidth-constrained AR streaming specifically.

“Existing video summarization patents and literature either rely on high-compute deep learning models (Transformers/LSTMs) unsuited for edge deployment, or use simplistic monaural audio volume spikes that fail to capture directional human interaction. Furthermore, current perceptual hashing techniques ignore human pose dynamics. The proposed PhysioSync-AR framework bridges this gap by coupling Inter-Channel Time Difference (ITD) spatial audio with sparse 3D skeletal landmark hashing, delivering real-time, privacy-preserving video summarization on standard CPUs.”

Expected outcome: Clarity of research gap and scope

By the end of the Problem Definition Stage, the team will have a precisely bounded research problem: existing summarizers either (a) use heavy Transformer/LSTM models unsuited to edge deployment, or (b) use monaural/volume-only audio cues and colour-only visual cues that miss directional, cross-modal events. The scope of PhysioSync-AR is fixed accordingly — a lightweight, CPU-only, cross-modal (spatial audio + skeletal pose) keyframe-hashing method for AR and lecture-capture video, explicitly excluding dense optical flow, Transformer-based retrieval, and non-spatial (mono) audio processing. This scoped problem statement and the identified literature/patent gap form the basis for the system design and implementation to be carried out in the next stage.